

Weaving Opportunities for Reflection into a Virtual Reality Learning Game

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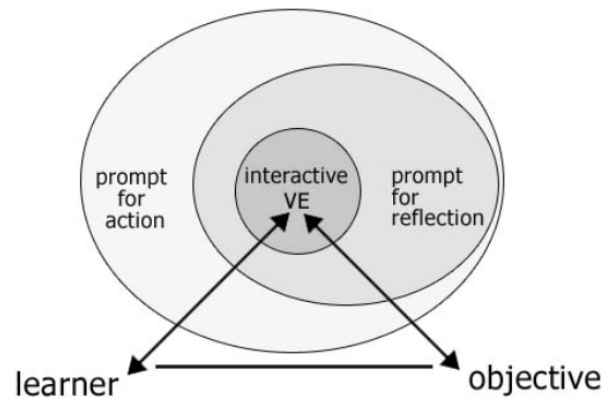


Figure 1: A suggestion to balance action and reflection in VEs for learning, presented as an Activity system diagram of Subject-Object-Tools.

ABSTRACT

This paper revisits and extends the Virtual Playground, a virtual environment originally developed two decades ago to explore the effects of interactivity in VR on children’s learning of abstract mathematical concepts. Initial findings from 2005 revealed that while interactive VR enhanced engagement and motivation, it sometimes hindered reflective processes critical to deep conceptual understanding. Recent advancements in VR, particularly affordable head-mounted displays and accessible development platforms, provided an opportunity to revive and modernize the Virtual Playground. The updated version, designed for the Quest headset, integrates structured reflection moments into the interactive experience to address the previously observed tension between interaction and reflection. We report preliminary work on embedding reflection opportunities in VR learning games and explore the design challenges encountered in balancing action and reflection. This work aspires to ignite a discussion about what makes immersive VR effective for learning, focusing on the interplay between interaction, reflection, and conceptual understanding.

Index Terms: Virtual Reality, conceptual learning, reflection

1 INTRODUCTION

Over the past decades, immersive virtual reality (iVR) has evolved significantly, offering new possibilities for educational applications. Early efforts in this domain explored how virtual environments (VEs) could enhance learning, with particular attention to abstract concepts and problem-solving skills. One such effort was the development of the *Virtual Playground*, an interactive VE aimed at studying

the role of interactivity in fostering conceptual understanding of mathematical concepts [10]. Initial empirical studies conducted in 2005 revealed that interactive VR experiences positively impacted children’s problem-solving abilities. However, they also highlighted a critical tension: while interactivity engaged learners and enhanced motivation, it often detracted from opportunities for deeper reflection [13]. Conversely, a “passive” VR version of the environment appeared to promote more reflective processes conducive to sustained conceptual change.

Considering these findings, future research directions were proposed, emphasizing the need to **balance interaction and reflection** in VR-based learning environments. However, technological limitations at the time curtailed further development of the Virtual Playground, as the system relied on high-end, now obsolete CAVE-based hardware. Recent advancements in VR technology, particularly the proliferation of affordable head-mounted displays (HMDs) and accessible development platforms like Unity™, have revived interest in revisiting this line of research. This paper reports on the preliminary work involved in reviving the Virtual Playground as a modern educational game. It explores the integration of reflection opportunities within an interactive VR learning experience.

The renewed Virtual Playground was reprogrammed to run on a Meta Quest 2 headset, transforming the original VE into a fully interactive educational game. In addition to technological updates, a key innovation in this revival was incorporating structured reflection moments inspired by the original study’s findings. These reflective opportunities were designed to balance the action-oriented nature of interactive VR with moments of cognitive pause to foster deeper conceptual understanding.

We begin by revisiting the initial empirical findings and design principles of the original Virtual Playground before detailing the process of reviving and redesigning the application for modern VR hardware. Next, we describe the reflective elements incorporated into the game, outlining our preliminary observations on how these elements influence learner engagement and problem-solving strategies. Finally, we discuss the ongoing challenges in designing for reflection in VR and propose future directions for research and

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development in this area.

2 BACKGROUND AND RELATED WORK

2.1 Reflection and Reflective Practices in Learning

Reflection has long been recognized as a critical component of meaningful learning, particularly in experiential and conceptual education. Boud, Keogh, and Walker [1] define reflection as an active process where learners revisit their experiences, critically analyze them, and derive meaning to foster deeper understanding. This foundational work underscores the importance of creating learning environments that encourage students to engage in reflective thinking, a practice that enhances metacognition and self-regulation.

Daudelin [2] further emphasizes reflection as essential for learning from experience. Learners can transition from mere experience accumulation to knowledge construction by integrating structured reflection prompts. Daudelin argues that reflection enables individuals to derive insights from complex, real-world tasks, making it particularly relevant in educational settings, prioritizing problem-solving and critical thinking.

Davis [3] introduces the concept of scaffolding reflection in science learning, proposing that guided reflection can bridge the gap between surface-level engagement and deep conceptual understanding. This perspective highlights the role of educators in facilitating reflective practices, ensuring that learners not only engage with content but also internalize and apply it effectively.

Schön [14] expands on the idea of reflective practice by distinguishing between reflection-in-action and reflection-on-action. Reflection-in-action occurs during the learning activity, enabling immediate adjustments, while reflection-on-action happens after the activity, allowing for retrospective analysis. This dual approach is especially relevant for immersive learning environments, where learners must continuously interpret and adapt to dynamic stimuli.

2.2 Immersive VR for Conceptual Learning: Challenges and Insights

The advent of iVR has opened new possibilities for experiential and conceptual learning. Immersive VR environments provide learners with high levels of presence and engagement, which are critical for fostering motivation and sustained interaction. However, research indicates that these benefits do not automatically translate to deeper conceptual understanding.

Roussou et al. [11, 13] provide contributions to understanding the role of interactivity in VR learning environments through their development of the Virtual Playground, a VR application designed to teach abstract mathematical concepts. Their findings highlighted a crucial tension: while interactivity enhanced learner engagement, it often hindered reflection, which is essential for deep conceptual change. This tension forms the primary motivation for the present study, which aims to address this balance by integrating reflective moments into interactive VR experiences.

Makransky et al. [8] explore the relationship between immersive VR and generative learning strategies. Their findings indicate that while VR increases learner enjoyment and presence, it may not inherently lead to better learning outcomes unless combined with effective instructional design elements such as reflection prompts and scaffolding. This insight underscores the need to balance engagement with cognitive processes that promote deeper learning. Earlier work by Makransky, Terkildsen, and Mayer [9] revealed that immersive VR could enhance learners' sense of presence but might inadvertently reduce learning when the experience becomes too stimulating, thus overwhelming cognitive load. The authors advocate for designing VR environments that manage cognitive load and integrate reflective elements to support conceptual change.

Lawson and Marchand Martella [6] critically review the current state of immersive VR in education, identifying gaps in understanding how VR can support reflection and deep learning. They call for

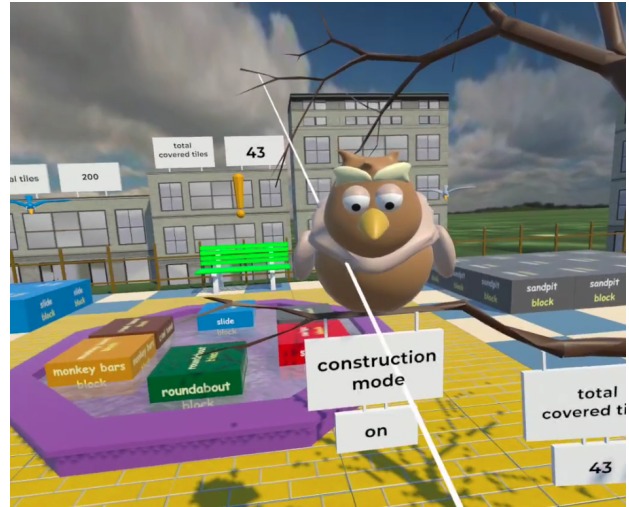


Figure 2: A virtual owl presents the rules and guides the users.

more empirical research on designing VR environments that promote not only immediate engagement but also long-term retention and conceptual change.

Hamilton et al. [4] conducted a systematic review of quantitative learning outcomes in VR-based education, concluding that immersive VR can enhance learning when appropriately designed but also cautioning that the novelty of VR alone does not guarantee improved outcomes. Wu, Yu, and Gu [15] examined the effectiveness of immersive VR using HMDs in education. Their results indicate that while VR shows promise in enhancing engagement and motivation, its impact on learning performance is highly variable, depending on the instructional design and integration of reflective practices.

2.3 Integrating Reflection in Immersive VR

Drawing from the above literature, a key design challenge in immersive VR is ensuring that high levels of presence and engagement do not come at the expense of reflection. Including reflective practices, such as scaffolding, prompts, and guided debriefing, can help learners internalize their experiences and foster deeper understanding.

Roussou et al. [12] suggest that immersive VR environments can benefit from “layers of prompts” that encourage learners to pause and reflect on their actions (Fig. 1). These prompts can take various forms, including visual cues, narrative elements, and intelligent agents. Laurillard’s [5] conversational framework provides a theoretical basis for designing such prompts, proposing that educational technologies should support discursive, adaptive, interactive, and reflective processes.

With the rise of generative AI, new opportunities have emerged to personalize reflection in VR environments. AI-driven agents can offer tailored feedback and adaptive prompts based on the learner’s actions, potentially enhancing both engagement and reflection. Lege and Bonner [7] highlight the potential of combining AI with immersive VR to create dynamic learning environments that respond to individual learner needs, promoting both immediate interaction and long-term reflection.

3 REVISITING THE VIRTUAL PLAYGROUND

3.1 Description of the original Virtual Playground

The Virtual Playground was developed in 2005 to investigate how interactivity in VR could influence children’s understanding of abstract mathematical concepts, particularly fractions. The learning task was

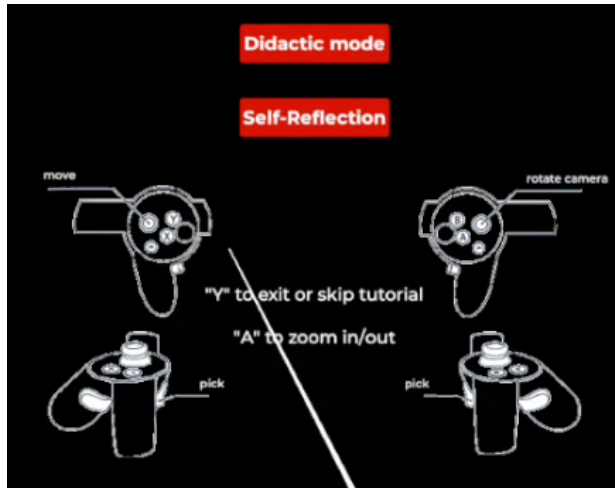


Figure 3: Interaction had to be remapped onto the HMD's controllers.

embedded within a playground-themed environment featuring six interactive elements: swings, monkey bars, a slide, a merry-go-round, a crawl tunnel, and a sandbox. These elements were color-coded and represented by blocks that needed to be correctly sized through fractional calculations. For instance, the task required learners to adjust the area of the playground elements by adding or removing blocks onto the playground tiles. To complete the task, they had to compare fractions (e.g., deciding whether $1/3$ or $1/4$ represented a larger amount) and apply their understanding to solve the problem. The VE provided intrinsic feedback, such as visual and audio cues, and employed a virtual owl and other agents to present rules and goals (Fig. 2).

Three experimental scenarios were tested with elementary school children, ages 8-11: interactive VR, passive VR, and a non-VR scenario using physical Legos. The interactive VR scenario allowed participants to manipulate virtual objects using a wireless wand in a fully immersive environment. The passive VR scenario involved observing a pre-recorded sequence where a virtual robot performed the tasks, encouraging participants to predict and explain its actions. The non-VR scenario used tangible Legos to replicate the tasks without digital mediation.

The findings revealed that while interactivity fostered engagement and problem-solving, it did not guarantee learning. In contrast, the passive VR scenario showed potential for promoting reflection, as participants observed and verbalized the robot's actions, engaging in reflective observation. This suggested that combining interactivity with guided prompts and reflective elements could enhance both engagement and understanding.

The insights from the original Virtual Playground experiments serve as a foundation for this paper's investigation into modernizing the VE by incorporating structured reflection moments.

3.2 Reviving and Redesigning the Virtual Playground

The original Virtual Playground was designed for high-end virtual reality systems, specifically CAVEs, powered by SGI Onyx2 Supercomputers and utilized standard CAVE application development software and open-source APIs such as OpenGL Performer. However, with the rapid evolution of technology, both the hardware and software became obsolete, particularly after SGI ceased operations in 2009. Consequently, reviving and extending this early work required a significant transition from projection-based VR systems to modern headset-based VR.

A fortunate aspect of the original development was that all 3D



Figure 4: Interaction with virtual characters, such as the birds, had to be reprogrammed to fit headset interaction

models were preserved in .obj format, enabling straightforward import into Unity. This allowed us to rebuild the environment by reprogramming the behavior of all virtual elements using Unity. Additionally, the interaction design had to be reimagined to fit the Meta Quest controllers, necessitating a complete remapping of control inputs (Fig. 3). Continuous locomotion, which was common in projection-based VR systems, had to be replaced with teleportation, a more suitable method for HMDs due to its ability to mitigate motion sickness.

Another key challenge was recreating the audio elements. In the original version, virtual characters' voices were recorded with actors in a studio, as reliable text-to-speech technology was not available at the time. For the updated version, we leveraged modern text-to-speech systems to generate the characters' voices, but we also kept some of the original audio samples (e.g., the birds' conveying the rules, see Fig. 4), as well as various environmental sound effects.

Finally, the introductory scenario narrated by the owl was revised and made more succinct, acknowledging that younger users have limited attention spans and may struggle with lengthy instructions. The owl now welcomes players with a concise opening line: "Welcome to the playground! Look around — things are a bit messy, with everything in odd sizes and places. Can you help us fix this? I'm here to guide you, so listen carefully!" before continuing with the rest of the scenario:

"First, click on me to switch to construction mode.

Now you're in construction mode! This view shows colored blocks representing each area on the playground. You'll use blocks to change size. You can click on me anytime to switch back and see your progress.

See that pool in the center? It's full of colored blocks you can use.

To make an area bigger, add blocks; to make it smaller, remove them. To pick a block, point to it in the pool and click the trigger. Move it to where you want to place it, then release the trigger to set it down. Each block is one tile, so remember, this is your unit.

In construction mode, you can also view the playground from above. Click on the 'A' button to try it and again to return to ground view.

Move close to each bird to learn what changes are needed. Only areas with a bird can be changed.

Switch to playground mode when you're ready to see if you sized things correctly. I'll check your work and give feedback if needed.

You can make changes in any order but remember: the total covered area must not exceed a quarter of the playground. Check the signs to see how many tiles you've covered.

Ready? Get set, go!"

3.3 Embedding reflection

3.3.1 Research questions guiding our approach

In attempting to incorporate reflection in the revived version of the Virtual Playground, we came up with the following research questions that span the design, integration within iVR (balancing with interaction), and evaluation of reflection in terms of its effectiveness:

- RQ1: How can opportunities for reflection be effectively embedded in interactive VR environments?
- RQ2: What are the design tensions and challenges in balancing interaction and reflection in VR learning games?
- RQ3: How do learners respond to reflection moments within an interactive VR game, and what impact does this have on their conceptual learning?

At this point of the design process, we have incorporated opportunities for reflection in the virtual reality game (RQ1) and have begun exploring RQ2. The next subsection describes the extensions we have implemented.

3.3.2 Guided instruction and self-reflection integrated into the game

Two modes were added to the original gameplay: Didactic Mode and Self-Reflection Mode.

Didactic Mode follows a guided instructional approach. The player interacts with the game as originally designed. However, after attempting to complete each task and seeing that it doesn't work, the option to get instructions is given by clicking on the virtual owl. An example of Didactic Mode follows:

"Hmm.. I am afraid that your answer is wrong but don't worry. I am here to help you understand this. Would you like that?"

A panel then appears with two buttons: 'Help Me' 'Solve It.'

- If 'Help Me' is selected, a sequence of steps resembling a class lesson is conveyed to the user.
- If 'Solve It' is selected, the system switches to the correct result without providing further explanation.

Self-Reflection Mode aims to leverage the motivational power of interactivity to engage learners in action while embedding support for reflection. This dual prompting—action and reflection prompts—can be integrated into various forms, such as audio-visual feedback, through intelligent agents and storytelling mechanisms that facilitate vicarious action. We plan to explore all these options in further application development.

For now, we have just implemented "pauses" after carrying out each task, whether the execution of the task is correct or not. Following each task, the system asks the user, "Do you want to see the answer or get time to think?" A panel appears with two buttons: 'Show Answer', 'I Need Time' (Fig. 5).

- If 'Show Answer' is selected, the user's answer is checked, and the mode is changed to reflect the result.
- If 'I Need Time' is selected, nothing happens, and the system responds: *"Ok! Take your time to think about it."*

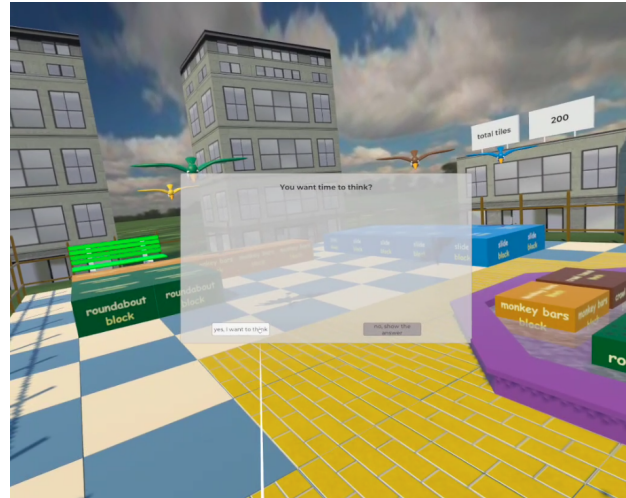


Figure 5: A user interface panel in self-reflection mode prompts the user to take time to think.

If the response is incorrect, the system provides feedback, such as: *"I am sorry, but your answer is wrong. Don't give up and try again!"* or *"Oops, not quite right! Don't worry; keep trying, you're getting closer!"*

While implementation has been completed, the above enhancements have not yet been evaluated with students. So far, usability evaluation has been conducted, and the virtual environment has been reviewed by three experts—two user experience designers and a teacher. As a result, detailed qualitative feedback was provided, which led to further design iterations currently underway. Once these iterations are complete, we aim to evaluate with students and collect learner responses to reflection opportunities. This will allow us to identify the challenges and successes in balancing interaction and reflection. We plan to take headsets to schools and evaluate the different modes with children in collaboration with teachers.

4 CONCLUSION AND FUTURE WORK

This paper aims to initiate discussion on the critical role of reflection in fostering deep learning, particularly in immersive VR environments where high engagement levels can overshadow reflective processes. Our hypothesis is that integrating reflective opportunities in interactive VR applications is crucial for enhancing conceptual learning. Effective VR learning environments should strike a balance between interaction and reflection, ensuring that learners participate actively and process their experiences meaningfully.

A preliminary attempt was made to insert opportunities for reflection into a fractions game for elementary-level children. Future research should focus on refining these reflective elements, on developing and testing more sophisticated design ideas that integrate reflective practices in VR by leveraging technologies like AI to create adaptive and personalized learning experiences, and on conducting empirical testing with learners and educators in school environments.

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